

Developing machine learning models to predict biogeochemical processes for the bioremediation of contaminated sites



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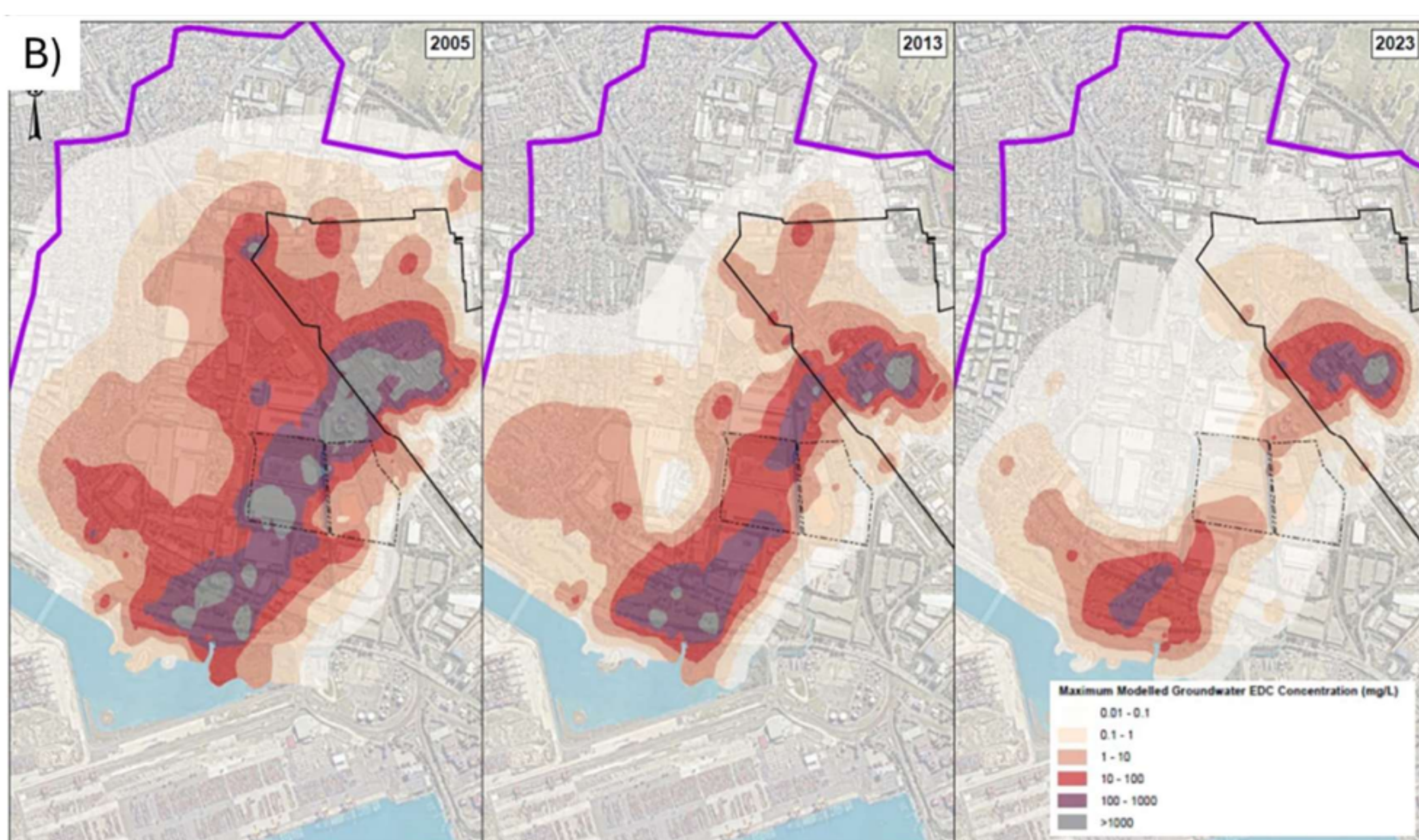
Motivation

Organohalides are pervasive environmental pollutants commonly found in contaminated soil and groundwater, causing serious harm to ecological stability and human health. Because traditional removal approaches pose a large economic and environmental burden, more sustainable practices have emerged, especially bioremediation.

Bioremediation enhances natural microbial communities, which can partially degrade organohalides such as organochlorines, by enriching the existing strains or inoculating new specialised ones. While long established, bioremediation has not yet benefitted from the recent advances in computational **predictive modelling** and microbiome data science.

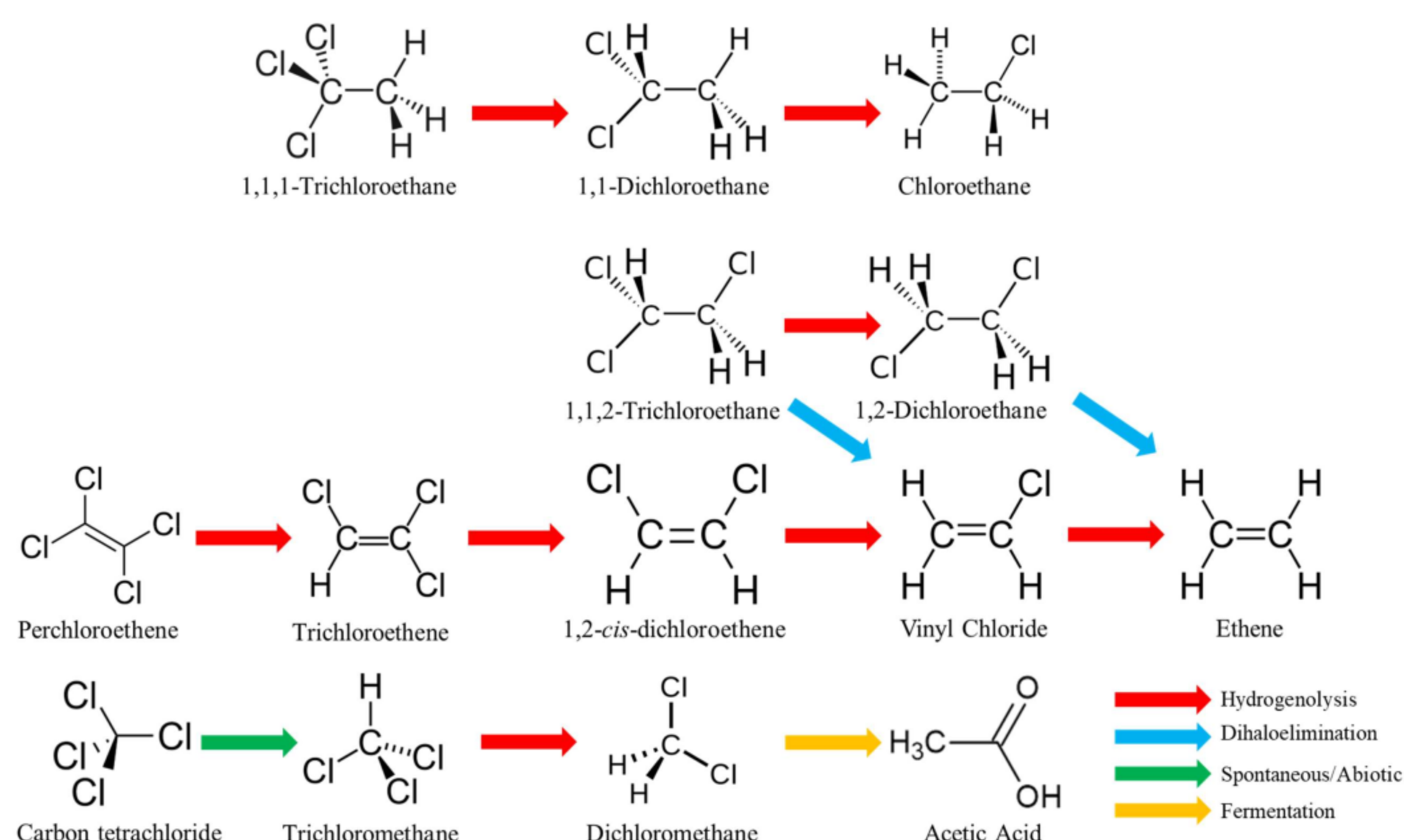
In this project, we will develop **machine learning** methods to predict *in-situ* organohalide biodegradation rates from microbiome profiles at a well-studied contaminated site in Australia, to ultimately reduce the uncertainty over bioremediation project performance.

Botany Industrial Park (BIP), Sydney



Reproduced from [1].

Organochlorine degradation pathways



Reproduced from [2].

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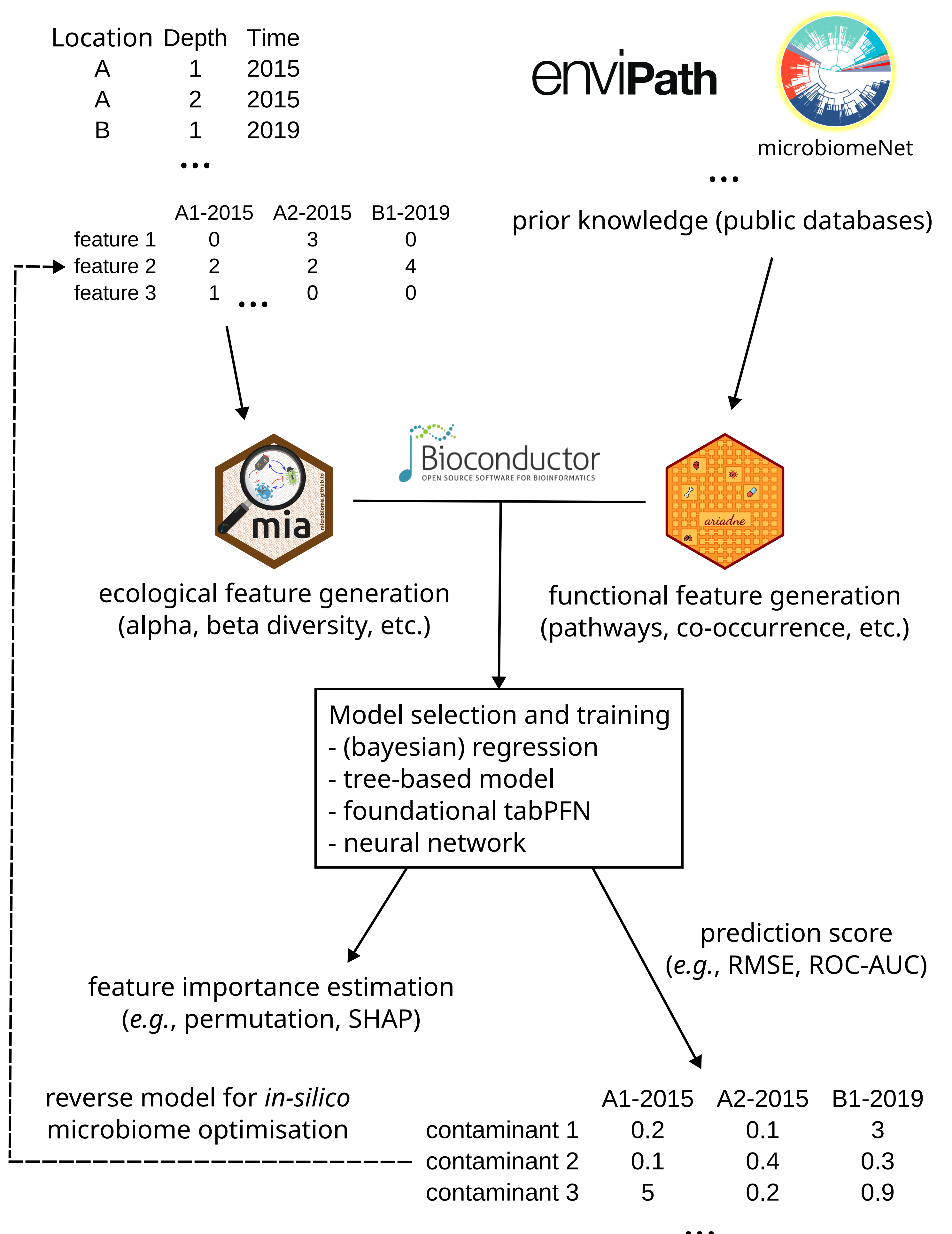
Data analysis

The **BIP data** includes groundwater chemical conditions, contaminant concentrations, and 16S rRNA gene sequencing data from multiple monitoring locations for over 20 years. Sample biochemical properties vary based on time, plume type, and collection depth. The time-series setup also allows for predicting future degradation trajectories.

Leveraging the **R/Bioconductor** ecosystem for microbiome data science [3], informative ecological features will be generated from the input counts table, enhanced with prior functional knowledge derived from selected public databases, such as metabolic pathways and microbial co-occurrence networks.

The generated features will be used to cross-validate different model categories (regression, tree-based, foundational and neural network) and predict contaminant **degradation rates** over time. For *in-silico* microbiome optimisation, a reversed pipeline could predict optimal microbial compositions based on contaminant concentrations.

Analytical pipeline



References

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3. Buorman, Tuomas, et al. "Orchestrating microbiome analysis with Bioconductor." *BioRxiv* (2025): 2025-10